

Prions

Prion diseases are caused by the conversion of a normal (predominantly α -helical) protein termed prion protein (PrP^c) to a β -pleated form (PrP^{sc}), which is transmissible via CNS-related tissue (iatrogenic CJD) or food contaminated by BSE-infected animal products (variant CJD). PrP^{sc} resists protease degradation and facilitates the conversion of still more PrP^c to PrP^{sc}. Resistant to standard sterilizing procedures, including standard autoclaving. Accumulation of PrP^{sc} results in spongiform encephalopathy and dementia, ataxia, startle myoclonus, and death.

Creutzfeldt-Jakob disease—rapidly progressive dementia, typically sporadic (some familial forms).

Bovine spongiform encephalopathy—also called “mad cow disease.”

Kuru—acquired prion disease noted in tribal populations practicing human cannibalism.

▶ MICROBIOLOGY—SYSTEMS

Normal microbiota:
dominant

Neonates delivered by C-section have microbiota enriched in skin commensals.

LOCATION	MICROORGANISM
Skin	<i>S epidermidis</i>
Nose	<i>S epidermidis</i> ; colonized by <i>S aureus</i>
Oropharynx	Viridans group streptococci
Dental plaque	<i>S mutans</i>
Colon	<i>B fragilis</i> > <i>E coli</i>
Vagina	<i>Lactobacillus</i> ; colonized by <i>E coli</i> and group B strep

Bugs causing food-
borne illness

S aureus and *B cereus* food poisoning starts quickly and ends quickly (exotoxin-mediated).

MICROORGANISM	SOURCE OF INFECTION
<i>B cereus</i>	Reheated rice. “Food poisoning from reheated rice? Be serious! ” (<i>B cereus</i>)
<i>C botulinum</i>	Improperly canned foods (toxins), raw honey (spores)
<i>C perfringens</i>	Reheated meat
<i>E coli</i> O157:H7	Undercooked meat
<i>L monocytogenes</i>	Deli meats, soft cheeses
<i>Salmonella</i>	Poultry, meat, and eggs
<i>S aureus</i>	Meats, mayonnaise, custard; preformed toxin
<i>V parahaemolyticus</i> and <i>V vulnificus</i> ^a	Raw/undercooked seafood

^a*V vulnificus* predominantly causes wound infections from contact with contaminated water or shellfish.

Bugs causing diarrhea

Bloody diarrhea	
<i>Campylobacter</i>	Comma- or S-shaped organisms; growth at 42°C
<i>E histolytica</i>	Protozoan; amebic dysentery; liver abscess
Enterohemorrhagic <i>E coli</i>	O157:H7; can cause HUS; makes Shiga toxin
Enteroinvasive <i>E coli</i>	Invades colonic mucosa
<i>Salmonella</i> (non-typhoidal)	Lactose ⊖; flagellar motility; has animal reservoir, especially poultry and eggs
<i>Shigella</i>	Lactose ⊖; very low ID ₅₀ ; produces Shiga toxin; human reservoir only; bacillary dysentery
<i>Y enterocolitica</i>	Day care outbreaks; pseudoappendicitis
Watery diarrhea	
<i>C difficile</i>	Pseudomembranous colitis; associated with antibiotics and PPIs; occasionally bloody diarrhea
<i>C perfringens</i>	Also causes gas gangrene
Enterotoxigenic <i>E coli</i>	Travelers' diarrhea; produces heat-labile (LT) and heat-stable (ST) toxins
Protozoa	<i>Giardia</i> , <i>Cryptosporidium</i>
<i>V cholerae</i>	Comma-shaped organisms; rice-water diarrhea; often from infected seafood
Viruses	Norovirus (most common cause in developed countries), rotavirus (↓ incidence in developed countries due to vaccination), enteric adenovirus

Common causes of pneumonia

NEONATES (< 4 WK)	CHILDREN (4 WK–18 YR)	ADULTS (18–40 YR)	ADULTS (40–65 YR)	ADULTS (65 YR +)
Group B streptococci	Viruses (RSV)	<i>Mycoplasma</i>	<i>S pneumoniae</i>	<i>S pneumoniae</i>
<i>E coli</i>	Mycoplasma	<i>C pneumoniae</i>	<i>H influenzae</i>	Influenza virus
	C trachomatis	<i>S pneumoniae</i>	Anaerobes	Anaerobes
	(infants–3 yr)	Viruses (eg, influenza)	Viruses	<i>H influenzae</i>
	C pneumoniae (school-aged children)		<i>Mycoplasma</i>	Gram ⊖ rods
	S pneumoniae			
	Runts May Cough			
	Chunky Sputum			
Special groups				
Alcohol overuse	<i>Klebsiella</i> , anaerobes usually due to aspiration (eg, <i>Peptostreptococcus</i> , <i>Fusobacterium</i> , <i>Prevotella</i> , <i>Bacteroides</i>)			
Injection drug use	<i>S pneumoniae</i> , <i>S aureus</i>			
Aspiration	Anaerobes			
Atypical	<i>Mycoplasma</i> , <i>Chlamydomphila</i> , <i>Legionella</i> , viruses (RSV, CMV, influenza, adenovirus)			
Cystic fibrosis	<i>Pseudomonas</i> , <i>S aureus</i> , <i>S pneumoniae</i> , <i>Burkholderia cepacia</i>			
Immunocompromised	<i>S aureus</i> , enteric gram ⊖ rods, fungi, viruses, <i>P jirovecii</i> (with HIV)			
Healthcare-associated	<i>S aureus</i> , <i>Pseudomonas</i> , other enteric gram ⊖ rods			
Postviral	<i>S pneumoniae</i> , <i>S aureus</i> , <i>H influenzae</i>			
COPD	<i>S pneumoniae</i> , <i>H influenzae</i> , <i>M catarrhalis</i> , <i>Pseudomonas</i>			

Common causes of meningitis

NEWBORN (0–6 MO)	CHILDREN (6 MO–6 YR)	6–60 YR	60 YR +
Group B <i>Streptococcus</i>	<i>S pneumoniae</i>	<i>S pneumoniae</i>	<i>S pneumoniae</i>
<i>E coli</i>	<i>N meningitidis</i>	<i>N meningitidis</i>	<i>N meningitidis</i>
<i>Listeria</i>	<i>H influenzae</i> type b	Enteroviruses	<i>H influenzae</i> type b
	Group B <i>Streptococcus</i>	HSV	Group B <i>Streptococcus</i>
	Enteroviruses		<i>Listeria</i>

Give ceftriaxone and vancomycin empirically (add ampicillin if *Listeria* is suspected; add acyclovir if viral encephalitis is suspected).

Viral causes of meningitis: enteroviruses (especially coxsackievirus), HSV-2 (HSV-1 = encephalitis), HIV, West Nile virus (also causes encephalitis), VZV.

In HIV: *Cryptococcus* spp.

Note: Incidence of Group B streptococcal meningitis in neonates has ↓ greatly due to screening and antibiotic prophylaxis in pregnancy. Incidence of *H influenzae* meningitis has ↓ greatly due to conjugate *H influenzae* vaccinations. Today, cases are usually seen in unimmunized children.

Cerebrospinal fluid findings in meningitis

	OPENING PRESSURE	CELL TYPE	PROTEIN	GLUCOSE
Bacterial	↑	↑ PMNs	↑	↓
Fungal/TB	↑	↑ lymphocytes	↑	↓
Viral	Normal/↑	↑ lymphocytes	Normal/↑	Normal

Infections causing brain abscess

Most commonly viridans streptococci and *Staphylococcus aureus*. If dental infection or extraction precedes abscess, oral anaerobes commonly involved.

Multiple abscesses are usually from bacteremia; single lesions from contiguous sites: otitis media and mastoiditis → temporal lobe and cerebellum; sinusitis or dental infection → frontal lobe.

Toxoplasma reactivation in AIDS.

Osteomyelitis

RISK FACTOR	ASSOCIATED INFECTION
Assume if no other information is available	<i>S aureus</i> (most common overall)
Sexually active	<i>Neisseria gonorrhoeae</i> (rare), septic arthritis more common
Sickle cell disease	<i>Salmonella</i> , <i>S aureus</i>
Prosthetic joint replacement	<i>S aureus</i> , <i>S epidermidis</i>
Vertebral involvement	<i>S aureus</i> , <i>M tuberculosis</i> (Pott disease)
Cat and dog bites	<i>Pasteurella multocida</i>
Injection drug use	<i>S aureus</i> ; also <i>Pseudomonas</i> , <i>Candida</i>

Elevated ESR and CRP sensitive but not specific.

Radiographs are insensitive early but can be useful in chronic osteomyelitis (A, left). MRI is best for detecting acute infection and detailing anatomic involvement (A, right). Biopsy or aspiration with culture necessary to identify organism.

Red rashes of childhood

AGENT	ASSOCIATED SYNDROME/DISEASE	CLINICAL PRESENTATION
Coxsackievirus type A	Hand-foot-mouth disease	Oval-shaped vesicles on palms and soles A ; vesicles and ulcers in oral mucosa (herpangina)
Human herpesvirus 6	Roseola (exanthem subitum)	Asymptomatic rose-colored macules appear on body after several days of high fever; can present with febrile seizures; usually affects infants
Measles virus	Measles (rubeola)	Confluent rash beginning at head and moving down B ; preceded by cough, coryza, conjunctivitis, and blue-white (Koplik) spots on buccal mucosa
Parvovirus B19	Erythema infectiosum (fifth disease)	“Slapped cheek” rash on face C
Rubella virus	Rubella	Pink macules and papules begin at head and move down, remain discrete → fine desquamating truncal rash; postauricular lymphadenopathy
<i>Streptococcus pyogenes</i>	Scarlet fever	Sore throat, C ircumoral pallor, group A strep, R ash (sandpaperlike D , from neck to trunk and extremities), L ymphadenopathy, E rythrogenic toxin, strawberry T ongue (SCARLET)
Varicella-zoster virus	Chickenpox	Vesicular rash begins on trunk E , spreads to face and extremities with lesions of different stages



Urinary tract infections

Cystitis presents with dysuria, frequency, urgency, suprapubic pain, and WBCs (but not WBC casts) in urine. Primarily caused by ascension of microbes from urethra to bladder. Ascension to kidney results in pyelonephritis, which presents with fever, chills, flank pain, costovertebral angle tenderness, hematuria, and WBC casts.

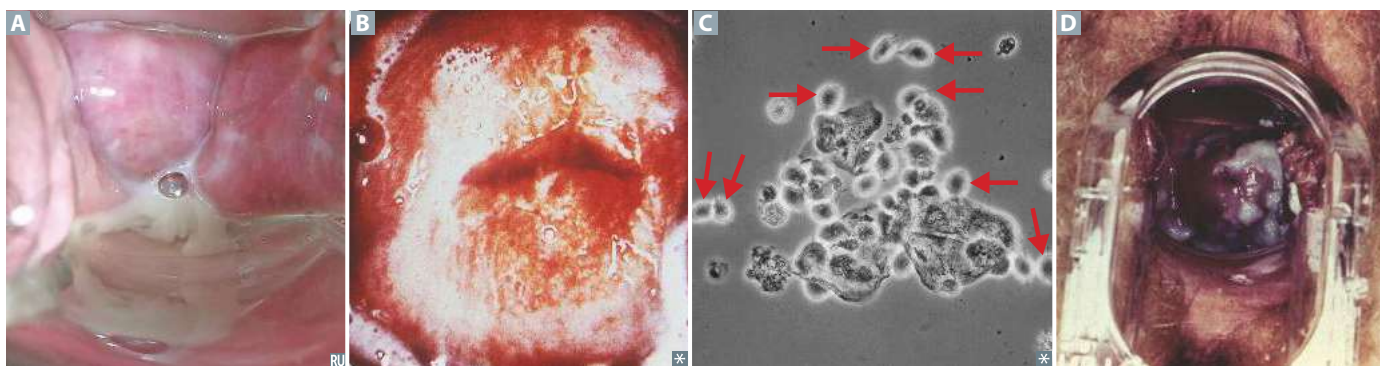
Ten times more common in females (shorter urethras colonized by fecal microbiota).

Risk factors: obstruction (eg, kidney stones, enlarged prostate), kidney surgery, catheterization, congenital GU malformation (eg, vesicoureteral reflux), diabetes, pregnancy.

SPECIES	FEATURES	COMMENTS
<i>Escherichia coli</i>	Leading cause of UTI. Colonies show strong pink lactose-fermentation on MacConkey agar.	Diagnostic markers: ⊕ Leukocyte esterase = evidence of WBC activity.
<i>Staphylococcus saprophyticus</i>	2nd leading cause of UTI, particularly in young, sexually active females.	⊕ Nitrite test = reduction of urinary nitrates by gram ⊖ bacterial species (eg, <i>E coli</i>).
<i>Klebsiella pneumoniae</i>	3rd leading cause of UTI. Large mucoid capsule and viscous colonies.	
<i>Serratia marcescens</i>	Some strains produce a red pigment; often healthcare-associated and drug resistant.	
<i>Enterococcus</i>	Often healthcare-associated and drug resistant.	
<i>Proteus mirabilis</i>	Motility causes “swarming” on agar; associated with struvite stones. Produces urease.	
<i>Pseudomonas aeruginosa</i>	Blue-green pigment and fruity odor; usually healthcare-associated and drug resistant.	

Common vaginal infections

	Bacterial vaginosis	<i>Trichomonas vaginitis</i>	<i>Candida vulvovaginitis</i>
SIGNS AND SYMPTOMS	No inflammation Thin, white discharge A with fishy odor	Inflammation B (“strawberry cervix”) Frothy, yellow-green, foul-smelling discharge	Inflammation Thick, white, “cottage cheese” discharge D
LAB FINDINGS	Clue cells (bacteria-coated epithelial cells) pH > 4.5 ⊕ KOH whiff test	Motile pear-shaped trichomonads C pH > 4.5	Pseudohyphae pH normal (4.0–4.5)
TREATMENT	Metronidazole or clindamycin	Metronidazole Treat sexual partner(s)	Azoles



Sexually transmitted infections

DISEASE	CLINICAL FEATURES	PATHOGEN
AIDS	Opportunistic infections, Kaposi sarcoma, lymphoma	HIV
Chancroid	Painful genital ulcer(s) with exudate, inguinal adenopathy A	<i>Haemophilus ducreyi</i> (it's so painful, you “do cry”)
Chlamydia	Urethritis, cervicitis, epididymitis, conjunctivitis, reactive arthritis, PID	<i>Chlamydia trachomatis</i> (D–K)
Condylomata acuminata	Genital warts B , koilocytes	HPV-6 and -11
Herpes genitalis	Painful penile, vulvar, or cervical vesicles and ulcers C with bilateral tender inguinal lymphadenopathy; can cause systemic symptoms such as fever, headache, myalgia	HSV-2, less commonly HSV-1
Gonorrhea	Urethritis, cervicitis, PID, prostatitis, epididymitis, arthritis, creamy purulent discharge	<i>Neisseria gonorrhoeae</i>
Granuloma inguinale (Donovanosis)	Painless, beefy red ulcer that bleeds readily on contact D Uncommon in US	<i>Klebsiella (Calymmatobacterium) granulomatis</i> ; cytoplasmic Donovan bodies (bipolar staining) seen on microscopy
Hepatitis B	Jaundice	HBV
Lymphogranuloma venereum	Infection of lymphatics; painless genital ulcers, painful lymphadenopathy (ie, buboes E)	<i>C trachomatis</i> (L1–L3)
Primary syphilis	Painless chancre F , regional lymphadenopathy	<i>Treponema pallidum</i>
Secondary syphilis	Fever, diffuse lymphadenopathy, generalized rash, condylomata lata	
Tertiary syphilis	Gummas, tabes dorsalis, general paresis, aortitis, Argyll Robertson pupil	
Trichomoniasis	Vaginitis, strawberry cervix, motile in wet prep	<i>Trichomonas vaginalis</i>



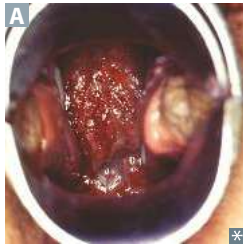
TORCH infections

Microbes that may pass from mother to fetus. Transmission is transplacental in most cases, or via vaginal delivery (especially HSV-2). Nonspecific signs common to many **ToRCHHeS** infections include hepatosplenomegaly, jaundice, thrombocytopenia, and growth restriction. Other important infectious agents include *Streptococcus agalactiae* (group B streptococci), *E coli*, and *Listeria monocytogenes*—all causes of meningitis in neonates. Parvovirus B19 causes hydrops fetalis.

AGENT	MATERNAL ACQUISITION	MATERNAL MANIFESTATIONS	NEONATAL MANIFESTATIONS
Toxoplasma gondii	Cat feces or ingestion of undercooked meat	Usually asymptomatic; lymphadenopathy (rarely)	Classic triad: chorioretinitis, hydrocephalus, and intracranial calcifications, +/- “blueberry muffin” rash A
Rubella	Respiratory droplets	Rash, lymphadenopathy, polyarthrits, polyarthralgia	Classic triad: abnormalities of eye (cataracts B) and ear (deafness) and congenital heart disease (PDA); +/- “blueberry muffin” rash. “ I (eye) ♥ ruby (rubella) earrings ”
Cytomegalovirus	Sexual contact, organ transplants	Usually asymptomatic; mononucleosis-like illness	Hearing loss, seizures, petechial rash, “blueberry muffin” rash, chorioretinitis, periventricular calcifications C CMV = C horio r etinitis, M icro c ep h aly, p eri V entricular calcifications
HIV	Sexual contact, needlestick	Variable presentation depending on CD4+ cell count	Recurrent infections, chronic diarrhea
Herpes simplex virus-2	Skin or mucous membrane contact	Usually asymptomatic; herpetic (vesicular) lesions	Meningoencephalitis, herpetic (vesicular) lesions
Syphilis	Sexual contact	Chancre (1°) and disseminated rash (2°) are the two stages likely to result in fetal infection	Often results in stillbirth, hydrops fetalis; if child survives, presents with facial abnormalities (eg, notched teeth, saddle nose, rhinitis, short maxilla), saber shins, CN VIII deafness



Pelvic inflammatory disease



Ascending infection causing inflammation of the female gynecologic tract. PID may include salpingitis, endometritis, hydrosalpinx, and tubo-ovarian abscess.

Signs include cervical motion tenderness, adnexal tenderness, purulent cervical discharge **A**.

Top bugs—*Chlamydia trachomatis* (subacute, often undiagnosed), *Neisseria gonorrhoeae* (acute).

C trachomatis—most common bacterial STI in the United States.

Salpingitis is a risk factor for ectopic pregnancy, infertility, chronic pelvic pain, and adhesions. Can lead to perihepatitis (**Fitz-Hugh–Curtis syndrome**)—infection and inflammation of liver capsule and “violin string” adhesions of peritoneum to liver **B**.

Healthcare-associated infections

E coli (UTI) and *S aureus* (wound infection) are the two most common causes.

RISK FACTOR	PATHOGEN	UNIQUE SIGNS/SYMPTOMS
Antibiotic use, PPIs	<i>C difficile</i>	Watery diarrhea, leukocytosis
Aspiration (2° to altered mental status, old age)	Polymicrobial, gram $\ominus$ bacteria, often anaerobes	Right lower lobe infiltrate or right upper/middle lobe (patient recumbent); purulent malodorous sputum
Decubitus ulcers, surgical wounds, drains	<i>S aureus</i> (including MRSA), gram $\ominus$ anaerobes (<i>Bacteroides</i> , <i>Prevotella</i> , <i>Fusobacterium</i>)	Erythema, tenderness, induration, drainage from surgical wound sites
Intravascular catheters	<i>S aureus</i> (including MRSA), <i>S epidermidis</i> (long term)	Erythema, induration, tenderness, drainage from access sites
Mechanical ventilation, endotracheal intubation	Late onset: <i>P aeruginosa</i> , <i>Klebsiella</i> , <i>Acinetobacter</i> , <i>S aureus</i>	New infiltrate on CXR, $\uparrow$ sputum production; sweet odor (<i>Pseudomonas</i>)
Exposure to blood products, shared medical equipment, needlesticks	HBV, HCV	
Urinary catheterization	<i>Proteus</i> spp, <i>E coli</i> , <i>Klebsiella</i> (PEcK)	Dysuria, leukocytosis, flank pain or costovertebral angle tenderness
Water aerosols	<i>Legionella</i>	Signs of pneumonia, GI symptoms (diarrhea, nausea, vomiting), neurologic abnormalities

Bugs affecting unvaccinated children

CLINICAL PRESENTATION	FINDINGS/LABS	PATHOGEN
Dermatologic		
Rash	Beginning at head and moving down with postauricular, posterior cervical, and suboccipital lymphadenopathy	Rubella virus
	Beginning at head and moving down; preceded by cough, coryza, conjunctivitis, and Koplik spots	Measles virus
Neurologic		
Meningitis	Microbe colonizes nasopharynx	<i>H influenzae</i> type b
	Can also lead to myalgia and paralysis	Poliovirus
Tetanus	Muscle spasms and spastic paralysis (eg, lockjaw, opisthotonus)	<i>C tetani</i>
Respiratory		
Epiglottitis	Fever with dysphagia, drooling, inspiratory stridor, and difficulty breathing due to edema	<i>H influenzae</i> type b (also capable of causing epiglottitis in fully immunized children)
Pertussis	Low-grade fevers, coryza → whooping cough, posttussive vomiting → gradual recovery	<i>Bordetella pertussis</i>
Pharyngitis	Grayish pseudomembranes (may obstruct airways)	<i>Corynebacterium diphtheriae</i>